

Limits and Velocity

MATH 145 – Applied Calculus

Torin Quinlivan

September 20, 2024



Instantaneous Rate of Change?

Recall in Lecture #1 we discussed the average rate of change:

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

This tells us the average rate of change across some interval $[a, b]$, but what if we want to know what the rate of change is at some point $(x, f(x))$.

Instantaneous Rate of Change?

Imagine you're on a roller coaster. How can you describe and predict how fast you'll be going at any particular point on the track?



Today we will be discussing the foundation of all calculus going forward: the **limit**.

$$\lim_{x \rightarrow a} f(x)$$

Everything we will do in future lectures: derivatives, integrals, all the things you may have heard of in terms of calculus are built off the humble limit.

Limits

Consider the function $f(x) = x^2$ at the point $x = 3$. As $f(x)$ is continuous, as we put x 's closer and closer to 3, we will get values of $f(x)$ closer and closer and closer to $f(3)$.

$$f(2) = 4$$

$$f(4) = 16$$

$$f(2.5) = 6.25$$

$$f(3.5) = 12.25$$

$$f(2.75) = 7.5625$$

$$f(3.25) = 10.5625$$

$$f(2.9) = 8.41$$

$$f(3.1) = 9.61$$

$$f(2.99) = 8.9401$$

$$f(3.01) = 9.0601$$

$$f(2.999) = 8.994001$$

$$f(3.001) = 9.006001$$

$$\vdots$$
$$\vdots$$

$$f(3) = 9$$

$$f(3) = 9$$

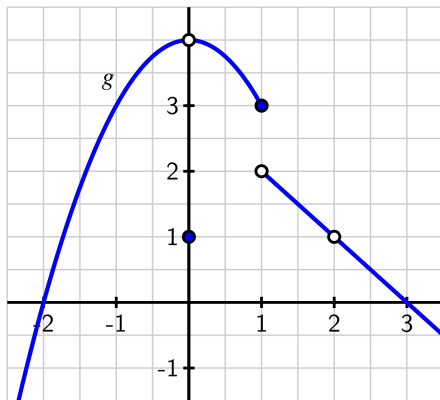
Limit

Given a function f , a fixed input $x = a$, and a real number L , we say that f **has limit** L as x **approaches** a , and write

$$\lim_{x \rightarrow a} f(x) = L$$

provided that we can make $f(x)$ as close to L as we like by taking x sufficiently close (but not equal) to a . If we cannot make $f(x)$ as close to a single value as we would like as x approaches a , then we say that f **does not have a limit** as x **approaches** a .

Limits on Graph Example



Example: Find:

■ $g(-1)$

■ $g(0)$

■ $g(1)$

■ $g(2)$

■ $\lim_{x \rightarrow -1} g(x)$

■ $\lim_{x \rightarrow 0} g(x)$

■ $\lim_{x \rightarrow 1} g(x)$

■ $\lim_{x \rightarrow 2} g(x)$

Left-Hand Limits

Left-Hand Limit

We say that f **has limit** L_1 **as** x **approached** a **from the left** and write

$$\lim_{x \rightarrow a^-} f(x) = L_1$$

provided that we can make the value of $f(x)$ as close to L_1 as we like by taking x sufficiently close to a , while always having $x < a$. We call L_1 the **left-hand limit of f as x approaches a** .

Right-Hand Limits

Right-Hand Limit

We say that f has limit L_2 as x approached a from the right and write

$$\lim_{x \rightarrow a^+} f(x) = L_2$$

provided that we can make the value of $f(x)$ as close to L_2 as we like by taking x sufficiently close to a , while always having $x > a$. We call L_1 the **right-hand limit of f as x approaches a** .

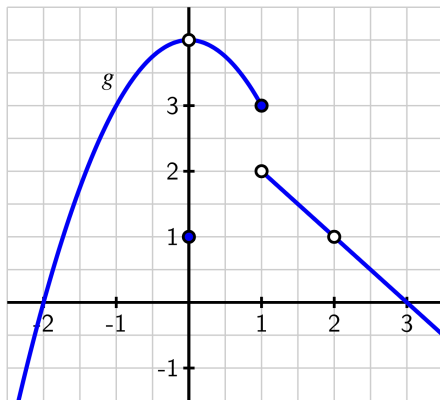
Left-Hand and Right-Hand Limits

Limit

A function f has limit L as $x \rightarrow a$ if and only if

$$\lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x)$$

Limits on Graph Example



Example: Find:

■ $g(-1)$

■ $\lim_{x \rightarrow -1^-} g(x)$

■ $\lim_{x \rightarrow -1^+} g(x)$

■ $\lim_{x \rightarrow -1} g(x)$

■ $g(1)$

■ $\lim_{x \rightarrow 1^-} g(x)$

■ $\lim_{x \rightarrow 1^+} g(x)$

■ $\lim_{x \rightarrow 1} g(x)$

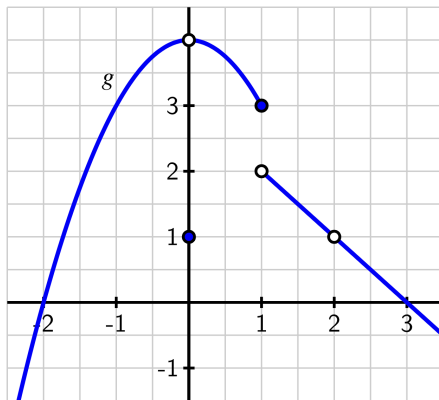
What does it mean for a function to be continuous?

Continuous at a Point

A function f is **continuous at** $x = a$ provided that

- f has a limit as $x \rightarrow a$,
- f is defined at $x = a$, and
- $\lim_{x \rightarrow a} f(x) = f(a)$.

Continuity Example



Example: Is f continuous at:

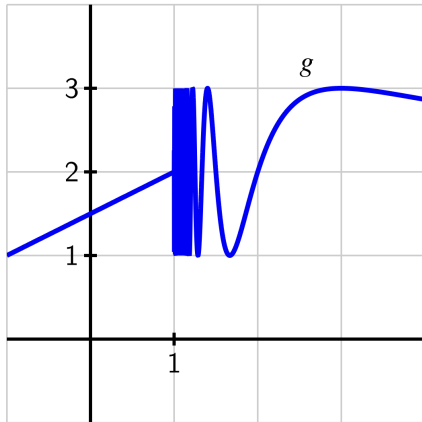
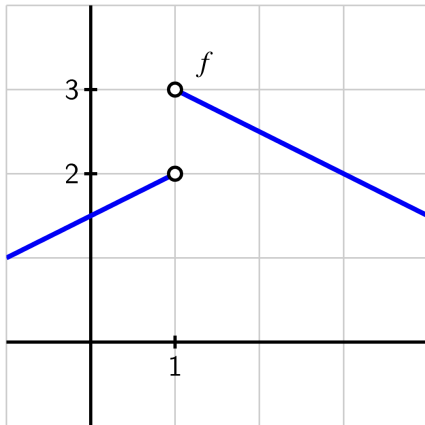
■ $x = -1$?

■ $x = 0$?

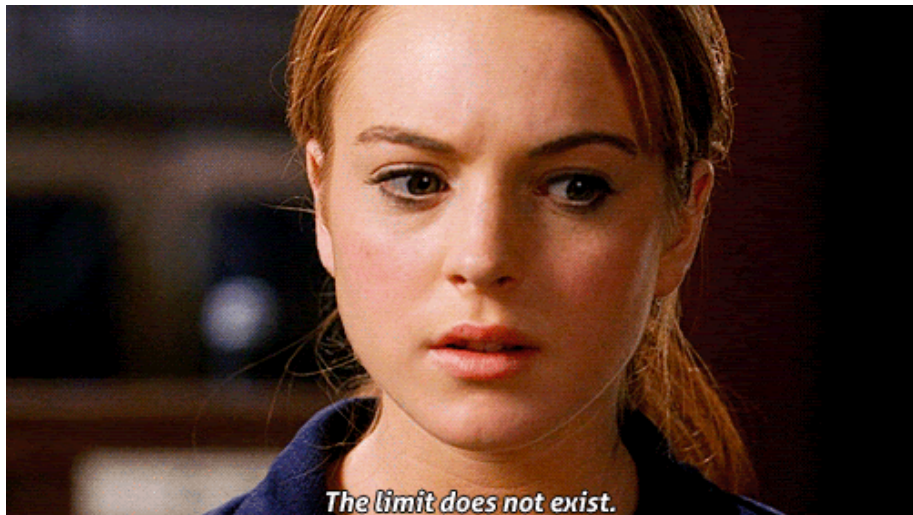
■ $x = 1$?

■ $x = 2$?

The Limit Does Not Exist!



The Limit Does Not Exist!



Limit Examples

Example: Try to find the following limits:

$$\lim_{x \rightarrow 1} x^2 + 2$$

$$\lim_{t \rightarrow 0} \frac{\sin(5t)}{t}$$

$$\lim_{y \rightarrow 1} \frac{x^2 - 1}{x - 1}$$

Rates of Change

Recall that:

Average Rate of Change

For a function f defined on an interval $[a, b]$, the **average rate of change** of f on $[a, b]$ is the quantity

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}.$$

Instantaneous Rate of Change

If we want to find the instantaneous rate of change at a point $x = a$, imagine the average rate of change over an interval that is getting smaller and smaller and smaller.

Instantaneous Rate of Change

For a function f , the **instantaneous rate of change** of f at point $x = a$ is the quantity

$$IV_{t=a} = \lim_{h \rightarrow 0} AV_{[a, a+h]} = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

Example: Consider $S(x) = x^2 + x + 1$.

- Find $AV_{[0,1]}$.
- Find $AV_{[-1,1]}$.
- Find $IV_{x=0}$.
- Find $IV_{x=1}$.